

Life Is Learned - Handoff 005 Audio Addendum

Packaged ElevenLabs Narration, Background Playback, and Local Fallback

Applies to: Handoff 005 - feature/handoff-005-library-platform

Baseline before H005: feature/handoff-004-6-six-stage-lessons at 74a18997dbf8931366ae5acd18f5a4a23ed8699d

Priority: Implement after 005A reader/content cleanup and before 005B/005C are considered frozen.

Reviewer: Clyde

Merge rule: Do not merge automatically. Push implementation + evidence and stop for reviewer approval.

1. Product decision

Life Is Learned now has a fourth production phase:

Research -> Authoring -> Illustration/Packaging -> Sound Production

Sound Production generates narration once with ElevenLabs, downloads the resulting audio, and packages it with the book. The shipping iOS app must **not call ElevenLabs at runtime** and must contain **no ElevenLabs credential**.

Playback precedence for an idea is strict:

1. If the exact installed idea revision has a **complete, valid packaged MP3 narration bundle**, use packaged audio for that entire idea.
2. If the idea does not have a complete valid bundle, use the existing on-device/local speech fallback for the entire idea.
3. Do **not** switch back and forth between ElevenLabs audio and local speech during one idea. Preflight the bundle before narration begins and choose one mode.
4. A transient network connection must never be required to listen to an already-installed narrated book.

This makes premium narration the default where available while preserving compatibility with every existing book.

2. Important implementation correction: an "idea MP3" is a bundle, not one monolithic file

Do not implement one single linear MP3 for a whole idea.

A Life Is Learned idea is interactive:

- six reading screens,
- two practice questions,
- learner-dependent feedback,
- a completion message,
- different Guide and Storyteller voices,
- pausing/replaying/page transitions,

- Story narration-follow scrolling.

A single continuous file would make practice branching, replay, page synchronization, and voice-role handling unnecessarily fragile.

Instead, treat **"MP3 available for the idea"** as shorthand for:

> every narration segment required to complete that idea is present, valid, and bound to the exact authored script.

The app still experiences this as one coherent premium narration mode.

3. Package contract - additive formatVersion 2 audio support

Keep formatVersion: 2. Add audio fields backward-compatibly.

Recommended types:

```
struct CollectionAudio: Codable, Sendable {
    var mediaType: String // audio/mpeg only in v1
    var data: Data // base64 in distribution JSON
    var durationMilliseconds: Int
}

struct NarrationCue: Codable, Sendable {
    var characterStart: Int
    var characterLength: Int
    var startMilliseconds: Int
    var endMilliseconds: Int
}

struct NarrationSegment: Codable, Identifiable, Sendable {
    var id: String
    var assetID: String
    var role: NarrationRole
    var scriptSHA256: String
    var durationMilliseconds: Int
    var cues: [NarrationCue]? // optional but preferred for synced highlighting/scrolling
}

struct NarrationProvenance: Codable, Sendable {
    var provider: String // "elevenlabs"
    var modelID: String?
    var guideVoiceID: String?
    var guideVoiceName: String?
    var storytellerVoiceID: String?
    var storytellerVoiceName: String?
    var producedAt: Date?
}

struct LessonNarrationBundle: Codable, Sendable {
    var schemaVersion: Int // 1
    var segments: [NarrationSegment]
    var provenance: NarrationProvenance?
}
```

Add to the package:

```
var audioAssets: [String: CollectionAudio]? = nil
```

Add to Lesson:

```
var narration: LessonNarrationBundle? = nil
```

Legacy books with no `audioAssets` or `narration` continue to decode and use local speech.

Do not overload the `image assets` table with audio. Artwork and audio have different validation, storage, and playback rules.

4. Required narration segments for a complete idea

The production pipeline must generate audio from the exact same strings the current narration engine speaks.

For each canonical six-stage idea, require:

Reading

One segment for each page:

```
page:<pageID>
```

The spoken script must match the current behavior:

```
<title>. <body text>
```

That is six page segments.

Questions

For each of the two questions:

```
question:<questionID>
```

The track contains the prompt plus every numbered answer option in the same order spoken by the existing narration system.

Feedback

Generate **every possible feedback branch**, not only the longest planning branch:

```
feedback:<questionID>:<choiceID>
```

The audio must include the same spoken prefix used by the app:

- correct: That 's right .
- incorrect: Let 's reconsider .

followed by that choice's authored feedback.

Completion

The existing completion narration includes first-try score. With exactly two questions, package the three possible variants:

```
completion:0  
completion:1
```

```
completion:2
```

Each maps to the exact completion string for 0/2, 1/2, or 2/2 first-try results.

If the product later removes the spoken score, the completion contract can become one static segment in a later revision.

5. Whole-idea completeness rule

Before playback begins, determine whether the idea has premium packaged narration.

A bundle is complete only when:

- every required segment ID exists exactly once,
- every segment references an existing MP3 asset,
- every MP3 decodes successfully,
- every segment's `scriptSHA256` matches the exact expected narration text,
- stored duration is positive and plausible,
- role matches the segment's expected Guide/Storyteller role,
- completion variants exist,
- every answer choice has its own feedback audio.

If **any** requirement fails, packaged narration is unavailable for that idea and the app uses local speech for the **whole idea session**.

Do not create a mid-idea voice switch because one clip is missing.

A corrupt premium bundle must never block reading, practice, progress, or card earning.

6. Script identity and revision semantics

This is important for learner progress.

Narration audio is a **presentation asset**, not new semantic lesson content.

Therefore:

- Changing lesson prose/questions/feedback still requires the normal idea revision increase.
- Changing only ElevenLabs voice, model, compression, loudness normalization, cue timing, or MP3 bytes while the `scriptSHA256` values remain identical requires a **collection revision**, but should **not** reset idea progress.
- The deterministic content comparison must not treat a voice-only re-render as a new pedagogical idea revision.
- If an audio segment's script hash does not match the current authored narration string, reject that premium bundle as stale.

This lets us improve narration later without making learners re-complete unchanged lessons.

7. Player architecture

Do not turn the existing `SpeechPlayer` into one giant class with two unrelated engines.

Preferred structure:

```
NarrationController
-> PackagedNarrationPlayer
-> LocalSpeechPlayer (existing AVSpeechSynthesizer behavior)
```

`NarrationController` chooses one mode per idea after bundle preflight and exposes the state the views already need:

- play
- pause
- resume
- stop
- isPlaying
- isPaused
- current narration segment
- current spoken/cue range when available
- completion callbacks

The existing `SpeechPlayer` implementation can become the local fallback implementation rather than being deleted.

The learner's existing Guide/Storyteller voice settings should be renamed/reframed as **device fallback voices**. They do not replace packaged studio narration when that narration is available.

Playback speed remains learner-controlled in both modes.

8. Packaged MP3 playback

Use file-backed `AVFoundation` playback, not repeated in-memory base64 decoding.

A suitable implementation is `AVPlayer` / `AVQueuePlayer` with one item per narration segment. Configure an audio time-pitch algorithm appropriate for spoken audio so learner speed changes preserve intelligibility.

At installation time, the Library-v3 package store may materialize validated MP3 assets to package-local files for efficient playback.

Do **not** keep two permanent copies of every MP3 (full embedded base64 JSON plus extracted MP3 files) if that doubles installed storage. The Library-v3 installer should normalize the distribution package into one durable installed payload representation.

Offload must delete those audio payload bytes along with prose/art payload while retaining cards, progress, favorites, and History.

9. Background and lock-screen playback

Packaged narration exists partly to solve the current lock-screen limitation.

Required behavior:

- Enable the iOS audio background mode for narrated playback.
- Use an `AVAudioSession` category suitable for spoken playback.
- Packaged narration may continue when the display locks or the app moves to background.
- Do **not** call `session.suspend()` merely because `scenePhase != .active` when packaged narration is actively playing.
- Genuine audio interruptions still pause/stop safely according to system expectations.
- Returning to the foreground must restore the correct page/segment state.
- Auto-run may continue through the reading screens while locked.
- Narration must stop and wait when interactive practice requires learner input.

Add basic Now Playing information for packaged narration:

- book title,
- idea title,
- current stage,
- book cover when available.

Support lock-screen Play/Pause. Additional remote commands are optional for H005 unless they are straightforward and well-tested.

The local fallback may retain its existing foreground limitations if background synthesis cannot be made reliable without destabilizing the app. Premium packaged narration is the required background path.

10. Story highlighting and narration-follow scrolling

Current local speech receives character ranges from `AVSpeechSynthesizerDelegate`. MP3 playback does not.

Do not use local TTS silently underneath premium MP3 just to obtain ranges.

Preferred solution: store cue timing in `NarrationSegment.cues`.

At minimum, cues should support sentence/paragraph-level synchronization sufficient to:

- highlight or identify the approximate current text region,
- keep Story narration-follow scrolling functional,
- avoid sudden jumps.

If the production tool can obtain word-level alignment, store it. If not, a deterministic coarse cue map may be derived from sentence word counts and the measured MP3 duration as a v1 fallback.

Missing cue metadata must **not** invalidate otherwise good MP3 narration. Audio still plays; the UI simply falls back to coarser progress/no word highlight rather than switching voices.

11. Audio validation and budgets

Keep the total encoded JSON package limit at **64 MiB** unless evidence proves it is too small.

Add audio-specific limits for authored releases:

- media type: audio/mpeg,
- speech-optimized mono MP3,
- target approximately **64 kbps** after production normalization,
- each individual audio segment ≤ 2 MiB,
- total decoded packaged audio target $\leq$ **40 MiB**,
- positive duration metadata,
- valid MP3 decode before import,
- no remote audio URLs inside the installed lesson package.

The existing 24 MiB image budget remains an image budget; audio does not count against it.

The overall 64 MiB distribution-package limit still applies after base64 expansion and all metadata.

Production should normalize loudness consistently across books/voices. Target spoken-word loudness around **-16 LUFS integrated** with safe true peaks; exact tooling may vary as long as the result is consistent and clipping-free.

12. Phase 4 production tooling

Add deterministic tooling under `Tools/` so premium narration can be produced and packaged without hand-editing JSON.

Recommended tools:

`Tools/export_narration_scripts.py`

Input: a reviewed authored book package.

Output: one manifest containing every required segment ID, exact script, role, and script SHA-256.

It must use the **same narration-string construction rules** as `LessonNarration / Tools/lesson_timing.py`.

`Tools/package_narration.py`

Input:

- reviewed book JSON,
- narration-script manifest,
- a directory of generated MP3 files,
- optional cue/timing data and production provenance.

Output:

- complete formatVersion-2 package with audioAssets and lesson narration manifests.

It must:

- verify every required clip,
- reject unknown/duplicate clips,
- compute/check script hashes,
- inspect MP3 decodability/duration,
- enforce audio budgets,
- never silently omit a segment,
- never modify lesson prose to fit audio.

Validator

Extend Tools/validate_package.py with an audio gate/report.

Report per idea:

- premiumNarration: complete | absent | invalid,
- segment count,
- audio bytes,
- total measured narration duration,
- voice/model provenance when supplied,
- any missing or stale segment IDs.

A book is still structurally importable with no premium narration. Premium audio is optional at the format level and preferred at playback.

13. ElevenLabs is production infrastructure, not runtime infrastructure

Our ChatGPT/ElevenLabs capability is part of **Sound Production**.

The production workflow is:

```
Reviewed authored package
-> export deterministic narration scripts
-> generate Guide/Storyteller speech with ElevenLabs
-> download the chosen masters once
-> normalize/validate MP3s
-> package audio + cues + provenance
-> run audio/package QA
-> import/download book normally
```

The iOS app never:

- stores an ElevenLabs API key,
- calls ElevenLabs to play a lesson,
- spends generation credits during learner playback,
- depends on internet access for an installed narrated book.

A downloaded MP3 is a durable book asset, just like an illustration.

14. Voice roles for production

Preserve the existing two-role editorial intent:

Guide

- older/mature male presentation,
- crisp and natural,
- calm authority without sounding like an announcer.

Storyteller

- younger female presentation,
- softer and more conversational,
- natural narrative cadence,
- avoid synthetic or overly dramatic delivery.

Do not hard-code a remembered ElevenLabs voice ID in source. Voice IDs used in production come from the connected ElevenLabs workspace and are recorded as provenance in the package/tooling output.

Once the production voices are approved, keep them stable across a batch unless the reviewer intentionally changes the audio style.

15. H005 integration changes

This addendum modifies H005 in the following places:

005A

Add packaged narration schema, player selection, audio validation, background playback, and fallback behavior after reader/content-contract cleanup.

005B - Library-v3 / Offload

Package storage accounting must include audio. Offload removes narration payload bytes. History/cards/progress retain no full MP3 data.

005C - Remote library

`package.bytes` and `checksum` naturally cover narration too. Discovery metadata may optionally expose:

```
"narration": {  
  "available": true,  
  "provider": "ElevenLabs"  
}
```

Do not make this field necessary for installation; the package remains authoritative.

005F / Settings polish

Clarify that voice pickers configure **device fallback voices** when studio narration is available.

16. Required tests and evidence

Add deterministic coverage for:

Package/model

- old packages with no audio still decode,
- complete audio bundle decodes,
- missing required segment makes premium narration unavailable but does not invalidate local playback,
- stale script hash is rejected for premium mode,
- invalid MP3 is rejected,
- audio-only rerender does not reset idea progress,
- prose change still requires idea revision,
- audio budgets are enforced.

Playback selection

- complete bundle -> packaged narration only,
- absent bundle -> local fallback only,
- incomplete bundle -> local fallback for the whole idea,
- never switch engines mid-idea,
- replay uses the same selected engine,
- speed changes apply to packaged narration.

Interactive practice

- both question prompt tracks play,
- every answer choice has the correct matching feedback clip,
- 0/2, 1/2, and 2/2 completion variants resolve correctly.

Background

On a physical iPhone:

- packaged narration continues after screen lock,
- packaged narration continues when app backgrounds,
- lock-screen Play/Pause works,
- foreground return restores correct stage,
- practice waits for learner input rather than auto-skipping,
- interruption handling does not accidentally advance an idea.

Story following

- cue metadata drives Story follow-scroll without local speech,
- missing cues degrade gracefully without changing narration engine.

Offload

- audio bytes contribute to displayed reclaimable size,
- offload removes them,
- reinstall/redownload restores premium narration without duplicating Idea Cards/progress.

Evidence location:

```
Evidence/Handoff005/PackagedNarration/
```

Include package fixtures, validator output, playback-mode tests, background-device notes/screenshots, and measured storage before/after offload.

17. Acceptance summary

This addendum is accepted when:

- ElevenLabs narration can be packaged once and played indefinitely offline,
- a complete premium bundle wins over local speech automatically,
- local speech is used only when the idea lacks a complete valid premium bundle,
- the app never mixes premium and local voices within one idea,
- interactive questions and feedback remain correct,
- Story auto-follow remains usable without running local TTS under the MP3,
- packaged narration continues through screen lock/background until learner interaction is required,
- narration audio can be offloaded and redownloaded with the book,
- voice-only audio updates do not reset learner progress,
- no ElevenLabs secret or runtime API dependency ships in the app.

After this lands, **Sound Production becomes a standard fourth phase for every newly produced book.**